

Practical Workbook
CT-361
Artificial Intelligence & Expert Systems



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LAB # 01

FACTS, RULES, AND QUERIES IN PROLOG

Q1. A person is healthy if they eat some food, and that food is a fruit.

SOLUTION:

```
% Q1
fruit(apple).
fruit(banana).

eats(ali,apple).
eats(saad,pizza).

healthy(Person):- eats(Person,Food),fruit(Food).
```

OUTPUT:

```
14 ?- healthy(ali).
true.

15 ?- healthy(saad).
false.
```

Q2. Two people are siblings if they have the same parent.

SOLUTION:

```
% Q2
parent(sohaib,ali).
parent(sohaib,saad).
parent(hamza,sohaib).

siblings(Person1,Person2):-
    parent(Parent,Person1),
    parent(Parent,Person2).
```

OUTPUT:

```
18 ?- siblings(ali,saad).  
true.
```

```
19 ?- siblings(ali,hamza).  
false.
```

Q3. Forbelow facts, rules, and queries write English Meanings.

- (1) color(carrots, orange).
- (2) likes(Person, carrots):-vegetarian(Person).
- (3) pass(Student) :- study_hard(Student).
- (4) ?- pass(Who).
- (5) ?- teaches(professor, Course).
- (6) enemies(X, Y) :- hates(X, Y), fights(X, Y).

SOLUTION:

1. Carrots are orange in color.
2. A person likes carrots if that person is a vegetarian.
3. A student passes if they study hard.
4. Who passes?
5. Which course does the professor teach?
6. X and Y are enemies if X hates Y and X fights Y.

Q4. Enter the program into Prolog and execute the queries shown below:

- Rule 1:** A person can ring a number **if** the person is located at a place, and that place has that phone number.
- Rule 2:** A person is at a location if they are directly **at** that place.
- Rule 3:** A person is at a location if they are **visiting** someone, and that someone is at that location.

Facts & Rules

```
ring(Person, Number) :- location(Person, Place), phone_number(Place, Number).  
location(Person, Place) :- at(Person, Place).  
location(Person, Place) :- visiting(Person, Someone), location(Someone, Place).
```

```
phone_number(rm303g, 5767).  
phone_number(rm303a, 5949).  
at(dr_jones, rm303g).  
at(dr_mike, rm303a).  
visiting(dr_mike, dr_jones).
```

Queries.

```
?- location(dr_bottaci, Pl).  
?- ring(dr_mike, Number).  
?- ring(Person, 5767).  
?- ring(Person, Number).  
?- ring(dr_jones, 999).
```

SOLUTION:

% Q4

```
ring(Person, Number) :- location(Person, Place), phone_number(Place, Number).  
location(Person, Place) :- at(Person, Place).  
location(Person, Place) :- visiting(Person, Someone), location(Someone, Place).
```

```
phone_number(rm303g, 5767).  
phone_number(rm303a, 5949).  
at(dr_jones, rm303g).  
at(dr_mike, rm303a).  
visiting(dr_mike, dr_jones).
```

OUTPUT:

```
22 ?- location(dr_bottaci, P1).  
false.
```

```
23 ?- ring(dr_mike, Number).  
Number = 5949 .
```

```
24 ?- ring(Person, 5767).  
Person = dr_jones .
```

```
25 ?- ring(Person, Number).  
Person = dr_jones,  
Number = 5767 .
```

```
26 ?- ring(dr_jones, 999).  
false.
```

Q5. For below english sentence write applicable Prolog facts, rules, and queries

(1) Maria reads logic programming book by author
peter lucas.

(2) Anyone likes shopping if she is a girl.

(3) Who likes shopping?

(4) kirke hates any city if it is big and crowded.

SOLUTION:

```
% Q5(1)
reads(maria,logic_programming_book).
author(peter_lucas,logic_programming_book).

% Q5(2)
likes(Person,shopping):-girl(Person).

% Q5(3)
%likes(Who,shopping).

% Q5(4)
hates(kirke,City):-big(City),crowdy(City).

girl(sara).
big(karachi).
crowdy(karachi).
```

OUTPUT:

```
28 ?- reads(maria,Book).
Book = logic_programming_book.

29 ?- likes(Who,shopping).
Who = sara.

30 ?- hates(kirke,City).
City = karachi.
```

Q6. For below english sentence write Prolog program. Facts, rules, and queries

- (1) jia is a woman.
- (2) john is a man.
- (3) john is healthy.
- (4) jia is healthy.
- (5) john is wealthy.
- (6) anyone is a traveler if he is healthy and wealthy.
- (7) anyone can travel if he is a traveler.

Queries.

- (1) Who can travel?
- (2) Who is healthy and wealthy?
- (3) traveler(john).

SOLUTION:

```
% Q6
woman(jia).
man(john).
healthy(john).
healthy(jia).
wealthy(john).

traveleer(Person):-healthy(Person),wealthy(Person).
can_travel(Person):-traveleer(Person).
```

OUTPUT:

```
34 ?- can_travel(Who).
Who = john .

35 ?- healthy(Who),wealthy(Who).
Who = john .

36 ?- traveleer(john).
true .
```

Q7. What answers do you get below queries for given program.

Program :

```
vegetarian(jose).  
vegetarian(james).  
vegetable(carrot).  
vegetable(egg_plant).  
likes(jose, X) :- vegetable(X).  
loves(Who, egg_plant) :- vegetarian(Who).
```

Queries :

```
?- vegetable(X).  
?- vegetable(potato).  
?- vegetarian(_).  
?- likes(jose, What).  
?- likes(Who, egg_plant).  
?- loves(Who, egg_plant).
```

SOLUTION:

```
% Q7  
vegetarian(jose).  
vegetarian(james).  
vegetable(carrot).  
vegetable(egg_plant).  
  
likes(jose,X):-vegetable(X).  
loves(Who,egg_plant):-vegetarian(Who).
```

OUTPUT:

```
77 ?- vegetable(X).
```

```
X = carrot ;
```

```
X = egg_plant.
```

```
78 ?- vegetable(potato).
```

```
false.
```

```
79 ?- vegetarian(_).
```

```
true .
```

```
80 ?- likes(jose, What).
```

```
What = carrot ;
```

```
What = egg_plant.
```

```
81 ?- likes(Who, egg_plant).
```

```
Who = jose.
```

```
82 ?- loves(Who, egg_plant).
```

```
Who = jose ;
```

```
Who = james.
```

Q8. What answers do you get below queries for given program.

Program :

```
father_of(joe,paul).
```

```
father_of(joe,mary).
```

```
mother_of(jane,paul).
```

```
mother_of(jane,mary).
```

```
male(paul).
```

```
male(joe).
```

```
female(mary).
```

```
female(jane).
```

Queries :

```
?- father_of(paul,mary).
```

```
?- father_of(X,mary).
```

```
?- mother_of(X,paul).
```

SOLUTION:

% Q8

```
father_of(joe,paul).  
father_of(joe,mary).  
mother_of(jane,paul).  
mother_of(jane,mary).  
male(paul).  
male(joe).  
female(mary).  
female(jane).
```

OUTPUT:

```
83 ?- father_of(paul,mary).  
false.
```

```
84 ?- father_of(X,mary).  
X = joe.
```

```
85 ?- mother_of(X,paul).  
X = iane.
```

Q9. What answers do you get below queries for given program.

SOLUTION:

```
% Q9
man(ahmed).
man(john).
man(ali).
woman(sara).
woman(fatima).
woman(amina).

parent(ahmed,sara).
parent(amina,ahmed).

married(sara,john).
married(ali,fatima).

brother(ahmed,ali).
brother(ali,ahmed).

mother(X,Y):-woman(X),parent(X,Y).

father(X, Y):-man(X),parent(X,Y).

sister(X,Y):-woman(X),parent(Z,X),parent(Z,Y),X\=Y.

brother(X,Y):-man(X),parent(Z,X),parent(Z,Y),X\=Y.

grandparent(X,Y):-parent(X,Z),parent(Z,Y).

uncle(X, Y):-brother(X,Z),parent(Z,Y).
```

OUTPUT:

```
92 ?- father(X,sara).  
X = ahmed .
```

```
93 ?- brother(X,ahmed).  
X = ali.
```

```
94 ?- parent(ahmed,Child).  
Child = sara.
```

```
95 ?- grandparent(GP,sara).  
GP = amina .
```

```
96 ?- uncle(U,sara).  
U = ali .
```

```
102 ?- woman(X).  
X = sara ;  
X = fatima ;  
X = amina.
```

```
103 ?- man(X).  
X = ahmed ;  
X = john ;  
X = ali.
```